

Supervisor Meeting Agenda

2 March 2026 — 30 minutes

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1. Progress Update (5 min)

- **DONE:** Task 1 — Unit tests and CI pipeline (all passing, GitHub Actions green)
- **DONE:** Task 2 — Linting, formatting, type checking
- **DONE:** Task 3 — Evidential actor integrated into SB3 (`EvidentialPPO`, `EvidentialActorCriticPolicy`, NIG distributions)
- **DONE:** Task 4 — Baseline configs (4 YAML overrides) + orchestration script (`run_experiment.py`) for ablation study
- **DONE:** PPO hyperparameter review — calibrated against literature (Andrychowicz et al. 2021, CaRL 2025, EPPO 2024)
- 154 unit tests passing, `make verify` green (lint + format + typecheck + tests)
- All code runs on CPU — zero GPU time used so far

2. Decisions Needed (15 min)

2.1 EKF Sensor Inputs (Task 7)

The agent’s observation comes from the EKF. The EKF can fuse any subset of the sensors below. Need to decide which combination to use — affects drift behaviour, covariance characteristics, and what is feasible on the Lemonworx vehicles.

Sensor	Provides	Drift?	Notes
Wheel odometry	Velocity, displacement	Accumulates	Always available on real vehicles
IMU	Angular vel., linear accel.	Gyro bias drift	Cheap, always available
GNSS (GPS/Galileo)	Absolute position	No drift (noisy)	Unreliable near buildings, unusable indoors
Camera (visual odom.)	Relative motion from features	Accumulates	Requires feature extraction; lighting-dependent
Ray-cast (sim. LiDAR)	Distance to surfaces	No drift	Requires scan-matching (e.g. Cartographer)
LiDAR	Point cloud	No drift	Dense; requires real hardware for deployment

Q: Which subset of sensors should the EKF fuse? What is available on the Lemonworx vehicles? What produces the most interesting covariance signal for the thesis without excessive implementation scope?

2.2 CARLA Map and Parking Scenarios (Task 5)

Current plan: use CARLA Town01 parking lot with 3 scenario types:

1. Perpendicular parking (90° spots, $2.5\text{m} \times 5\text{m}$)
2. Parallel parking (street-side, $2.5\text{m} \times 6\text{m}$)
3. Angled parking (45° spots, $2.5\text{m} \times 5.5\text{m}$)

Q: Should I use a different or larger map? Should I start with one scenario type or implement all three from the beginning?

Q: Forward-entry only (current action space has no reverse gear) — is this acceptable?

2.3 Obstacle Perception

Without perception, the agent memorises the training layout and cannot detect novel obstacles. Some form of spatial awareness is needed so the agent can generalise across layouts.

Option	Observation	Weather affects it?	Scope
Ray-cast / ultrasonic	4–8 distance floats	No	Minimal
LiDAR (sector-reduced)	8–16 min-distance floats	Yes (fog, rain)	Moderate
LiDAR (full point cloud)	Thousands of 3D points	Yes	Significant
Camera (RGB)	Images ($640 \times 480 \times 3$)	Yes (fog, glare)	Massive

Q: Which perception modality? Weather-immune sensors (ultrasonic/ray-cast) isolate the localisation uncertainty variable cleanly. Weather-dependent sensors (LiDAR/camera) introduce a confounding variable but are more realistic.

2.4 Teleoperation Handoff Evaluation (Task 11)

The thesis claims the agent “drives more carefully when uncertain”. One safety mechanism is handing off to a human teleoperator when evidential uncertainty exceeds a threshold.

Q: How should I evaluate handoff outcomes? Switching to manual control is not necessarily bad — the model might also handle high uncertainty by driving cautiously on its own. What should the comparison look like?

2.5 Compute Resources and Seeds

Full ablation study: 4 baselines $\times$ N scenarios $\times$ N seeds.

Per-seed time estimate (assumptions below):

Parameter	Value
Timesteps per seed	1M
Episode length	100–300 steps (parking is short; max 500)
CARLA sync tick rate	50–200 steps/sec (bottleneck is CARLA, not policy)
Scene	Single parking lot, low-poly, vector obs only (no camera rendering)
Weather + NPCs	Randomised per episode; 0–5 vehicles, 0–3 pedestrians
Est. per seed	3–6 hours

Parallelisation strategy — two levels available:

1. **Run-level parallelism (implemented):** Multiple independent training runs execute simultaneously, each with its own CARLA instance on a separate port. Seeds are embarrassingly parallel — no coordination needed. Controlled via `parallel_workers` in config or `--workers` CLI flag.

Pros: Simple, already implemented, no impact on training dynamics.

Cons: Each worker needs its own CARLA instance ($\sim 2\text{--}4$ GB VRAM each).

2. **Env-level parallelism (not implemented, optional):** Multiple CARLA environments feed into a *single* PPO model within one training run (SB3 `SubprocVecEnv`). The rollout buffer fills faster because data is collected from multiple envs simultaneously.

Pros: Each individual run finishes faster.

Cons: Doubles CARLA instances needed per run; changes training dynamics (PPO hyperparameters like `n_steps` and `batch_size` were calibrated for a single env); more complex to implement.

Current state: Run-level parallelism is implemented and ready. Env-level parallelism is not needed unless individual runs prove too slow.

Seeds	Total runs	1 worker	2 workers	3 workers
5	20	3–5 days	1.5–2.5 days	1–2 days
10	40	5–10 days	2.5–5 days	2–3 days
15	60	8–15 days	4–7.5 days	3–5 days

Note: parking-only scenes with vector observations (no camera rendering) are lightweight. Each CARLA instance needs $\sim 2\text{--}4$ GB VRAM, so a 24 GB GPU can run 2–3 instances in parallel. Actual throughput depends on GPU model.

Q: What GPU hardware is available to me and what do I need to do to get access? How many parallel CARLA instances can the hardware support? Is 5 seeds acceptable if compute is limited?

3. Timeline and Next Steps (5 min)

Task	Est. Time	Blocked on
Task 6: Docker stack verification	1–2 days	GPU machine
Task 5: CARLA parking environment	1 week	Map decision (today)
Task 7: EKF filtered observation	3–5 days	Sensor decision (today)
Task 8: End-to-end training	2–3 days	Tasks 5, 6, 7
Task 9: Reward function tuning	1–2 weeks	Task 8
Training runs (4 baselines $\times$ 10 seeds)	2–4 weeks	Task 9
Tasks 10–14: Evaluation + analysis	2 weeks	Trained models
Dissertation writing	Ongoing	Can start now

Q: Does this timeline align with the submission deadline? Any priorities to adjust?

4. Dissertation Writing (5 min)

Starting to write chapters that do not depend on results:

- Introduction — problem statement, contributions
- Background — evidential deep learning, EKF localisation, PPO, uncertainty in RL

- Methodology — 2×2 ablation design, architecture, observation space
- Experimental setup — CARLA environment, Docker stack, evaluation conditions

Q: Any guidance on dissertation structure, expected length, or formatting requirements?